

CSI247: Lab 2 – Objects and Generics (7 am)

At the end of this lab you should be able to:

- Declare interfaces
- Implement an interface
- Implement abstract classes
- Implement inheritance
- Implement the Comparable interface
- Overriding super class method.
- Follow proper java project folder structure and compile from the command line.

Lab Question

In this lab, you will be creating classes that use generics as follows:

- Create a project folder called *lab2*
- Create the src and classes directories inside your projec root.

Polygon Interface

- Create a package *csi247.lab2.polygon*
- Create an interface called *Polygon*.
- Declare a method called *area()* that returns a double.
- Declare a method called *perimeter()* that returns a double.

Triangle Abstract Class

- Create an abstract class called *Triangle* in the *csi247.lab2.polygon*.
- The *Triangle* class should implement the *Polygon* interface as well as the *Comparable* interface.
- Create protected instance variables to represent the triangle sides.
- Create a constructor that takes parameters that match the instance variables.
- Implement the *perimeter* method.
- Implement the *compareTo* method to compare using the perimeter.

EquilateralTriangle

- Create an class called *EquilateralTriangle* in the *csi247.lab2.polygon*.
- The class should inherit from the *Triangle* class
- Implement a constructor that takes only one parameter and use it to set all the 3 sides of a rectangle.
- Implement the *area* method.
- Override the *toString* method

Tester Class

- Create a class called *Tester* in the *csi247.lab2* package.
- Create the main method.
- Declare a *Polygon* variable.
- Create an *EquilateralTriangle* object and assign it to the *Polygon* variable.
- Print the area and perimeter.
- Declare a *Triangle* variable and assign the object held by the *Polygon* variable.
- Use functional method implementation to compare triangles by their sides.
- Use a functional method implementation to compare the triangles by their areas.
- Write the code to test the functional methods.
- Print the value from the *toString()* method.